PROJECT MAP · ARXIV 2605.06527 · AS OF 12 OCT 2026

Does Write-Side State Adjudication Survive Small Models?

Reproducing and stress-testing STALE and CUPMem. Every branch of the map is a panel; every item carries a line of context explaining what it means and why it matters.

🟢 done 🟠 to do 🔴 open question | Read the left column for **what we are reproducing and how** , the right column for **what we have built and learned** . Panels are numbered in reading order.

Can a cheap model reliably decide which memories are still true?

STALE / CUPMem reproduction

`Reproduction + stress test · MLRC / TMLR · ACL/EMNLP workshops`

Assistants keep long-term memories about their users. New information can make an old memory false **without ever contradicting it**, and the STALE benchmark shows most models miss this.

The paper's fix, **CUPMem**, decides what is still true at the moment memory is written. It scores 68.0%, the best on STALE. We check whether that result reproduces, and whether it still holds when the deciding model is small and cheap.

68.0%	55.2%	8.7%
CUPMEM, REPORTED	BEST FRONTIER MODEL	GPT-4O-MINI B0

01 Source paper

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Chao et al., STALE, arXiv 2605.06527 (May 2026). Code (MIT) and data (CC BY 4.0) are both public, including the full CUPMem implementation.

- **STALE: can agents tell when memories are no longer valid?**
  A long-term assistant stores facts about its user. The benchmark asks whether it notices when one of those facts has quietly stopped being true.
- **Implicit conflict: a later observation invalidates an old memory with no explicit negation**
  Example from the data: a user mentioned a relationship, and much later talks about speed-dating. Nobody says "we broke up"; the agent has to infer it.
- **400 scenarios, 1,200 queries, contexts up to ~150K tokens**
  Three queries (probes) per scenario, asked over a history far too long to skim. Our own count: 151.7K tokens on average, 164.9K max.
- **Probes: SR state resolution · PR premise resistance · IPA policy adaptation**
  SR asks what is true now. PR slips the outdated fact into the question as a premise to see if the model goes along with it. IPA checks whether its advice adapts to the new state.
- **Best frontier model 55.2% overall; GPT-4o-mini B0 8.7%**
  Even the strongest model fails almost half the time. 8.7% is the number our B0 run must match (Table 2: T1 30.0 / 0.0 / 11.0, T2 9.5 / 0.0 / 1.5).
- **CUPMem: write-time state consolidation + propagation-aware search, 68.0%**
  The paper's own method and the best score on STALE. This is the headline number RQ1 tries to reproduce.

02 Dataset

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STALEproj/STALE on Hugging Face (HF sha 617c51dc). Every figure below was verified on our local copy.

- 🟢 `T1_T2_400_FULL.json` **, 306 MB, CC BY 4.0**
  The whole benchmark is one file. The licence lets us publish results and derived subsets with attribution.
- 🟢 **200 Type I + 200 Type II instances, 50 sessions each**
  The two types are scored separately. In the smoke test they failed differently: Type I answers reaffirmed the old state, Type II answers claimed the history never mentioned it.
- 🟢 **Gold evidence:** `[idx_old, idx_new]` **, about 27 sessions apart**
  Two sessions matter per instance: the one that set the old state and the one that silently invalidated it. The other 48 are distractors.
- 🟢 **Mean 151.7K tokens (max 164.9K); every instance exceeds 128K**
  GPT-4o-mini's window is 128K, so B0 always has to truncate. This matches the asterisk on GPT-4o-mini in the paper's Table 2.

03 Evaluation protocol

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Taken from the official code rather than the paper text, so our scores are directly comparable to theirs.

- 🟢 **One joint judge call per instance, temperature 0, JSON mode**
  The judge sees all three answers for an instance at once and returns a verdict for each. Fixed settings keep grades stable across reruns.
- 🔴 **Judge: Gemini-3.1-flash-lite (exact model id unconfirmed)**
  The paper names the model but not the API id string. A different snapshot could shift every score, so this needs confirming before E1.
- 🟢 **Overall = unweighted mean of 6 (type × dimension) cells**
  2 types × 3 probes. Each cell counts equally however many instances it has; `eval/stats.py` computes it the same way.
- 🟢 **Truncation to 128K: head ≤2, tail ≤2, then random between evidence sessions**
  Reserve 2,048 + 512 tokens for the answer, then drop sessions in that order until the history fits. The two evidence sessions are never dropped. In practice 8–11 sessions go per request.

04 Research questions

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Ordered so each builds on the last: the stress tests only mean something once RQ1 has landed.

- **RQ1 Reproduction: paper vs official code vs our reimplementation**
  Because the code is public, this is a three-way comparison. Any gap can be pinned on the paper's description, the code, or our rebuild.
- **RQ2 Small 3–9B open adjudicators**
  CUPMem uses an LLM to decide, at write time, what is still true. This tests whether cheap open models can make those calls or whether the method needs a frontier model. It is the paper's title question.
- **RQ3 Stage-wise failure attribution**
  When an answer is wrong, locate the stage that failed: the write-time decision, the propagation of the invalidation, retrieval, or the final answer.
- **RQ4 Ablations and cost**
  Remove components one at a time and measure tokens and dollars per instance. The paper reports about $0.37 per instance for CUPMem on GPT-4o-mini.
- **RQ5 2-hop cascades**
  STALE's conflicts are one step deep. In a 2-hop cascade the invalidated fact implied a second fact, which should now be dropped too. TEPA's authors call multi-hop out of scope, so this is open ground.

05 Systems under test

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Everything below runs through the same data, prompts and judge, so differences in score come from the system itself.

- 🟢 **B0 full context + evidence-preserving truncation**
  GPT-4o-mini reads the whole history, minus the sessions truncation removes. The baseline that every memory system has to beat.
- 🟠 **Official CUPMem wrapper (reference run)**
  The authors' own code, run by us. It must be run in both session modes, because the default mode only reads the gold sessions (see Open questions).
- 🟠 **M independent CUPMem reimplementation**
  Written from the paper, not the code. It tests whether the paper says enough to rebuild the method. Lives in `memory/cupmem/` .
- 🟠 **B1 naive RAG**
  Embed the sessions and retrieve the closest ones for each probe. Shows how much of CUPMem's gain retrieval alone would give.
- 🟠 **B2 Mem0**
  A widely used open-source memory layer, standing in for what practitioners deploy today.
- 🔴 **StateAuditor as a possible readout-side baseline**
  Checks memory when it is read, not when it is written: a direct contrast for a paper about write-side adjudication. Not yet decided.

06 Related work

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Recent papers on stale memory, checked 27 Sep 2026. Neither covers what RQ2–RQ5 ask.

- **StateAuditor (2608.01619): readout-side audit, +5.0 on STALE**
  Sun & He. Runs on full STALE but does not touch CUPMem, small adjudicators or cascades. Candidate baseline (see Systems under test).
- **TEPA (2608.07429): revokes keyed precedents; flags multi-hop as out of scope**
  Zhou et al. Evaluated on MemoryAgentBench-style settings, not STALE. Its stated limit on multi-hop is the gap RQ5 fills.

07 Harness

7

`src/stale_repro/` *: the shared code every system runs through. Offline tests use a scripted fake LLM, so they need no API keys.*

- 🟢 `data/loader.py` **records → Instance/Session/Probe, stratified subsets**
  Turns the 306 MB JSON into typed objects and draws deterministic, type-balanced subsets, such as the 50-instance dev set (25 per type).
- 🟢 `llm/gateway.py` **sqlite cache, JSON decoding, usage, TPM limiter**
  Every LLM call goes through here. The cache makes reruns free and identical; the limiter was added after we hit the 200K tokens/min account cap.
- 🟢 `memory/base.py` **MemorySystem interface + official prompts**
  One interface for every system, so B0, B1, B2 and CUPMem swap in by config. Response prompts are copied verbatim from App. E.1.
- 🟢 `eval/judge.py` **official joint judge, verbatim prompt**
  Uses the authors' judge prompt word for word, so our grades mean the same thing as theirs.
- 🟢 `eval/stats.py` **6-cell mean, bootstrap CI, McNemar**
  Overall is computed exactly as in the paper. Bootstrap gives error bars; McNemar tests whether two systems really differ on the same instances.
- 🟢 `runner.py` **config-driven, resumable, parallel,** `--no-judge`
  One YAML per experiment, and interrupted runs pick up where they stopped. `--no-judge` lets us generate answers now and grade them once a Gemini key arrives.
- 🟠 **Diagnostics / attribution module**
  Needed for RQ3. It will read the per-probe traces the runner already saves in `answers.jsonl` .

08 Findings so far

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From the first API runs on 3 Oct 2026 (B0, GPT-4o-mini).

- 🟢 `gpt-4o-mini-2024-07-18` **still served, pinned**
  The `gpt-4o-mini` alias resolves to this snapshot. Pinning it means a silent model update cannot change results mid-project, and no substitute model is needed.
- 🟢 **200K TPM limit ≈ 1 B0 call/min: dev 2.5 h, full E1 ≈ 20 h**
  Each call sends ~123K prompt tokens, and OpenAI counts ~148K because it reserves 16,384 output tokens. Runtime, not money, is the bottleneck.
- 🟢 **Truncation parity: estimate within 2 tokens of billed**
  123,209 estimated vs 123,211 billed. Our truncation keeps exactly the sessions the official code would.
- 🟢 **Smoke test: 4/4 instances, 1.49M input tokens, ≈ $0.23**
  12 real calls end to end. With the limiter in place the resumed run hit no rate-limit errors.
- 🟢 **PR answers accept false premises; SR reaffirms old state**
  E.g. planning anniversary dates for a user who has since started speed-dating. Consistent with the paper's near-zero PR for GPT-4o-mini; not yet graded by the judge.

09 Open questions

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Places where the paper and the code disagree or say nothing. Each needs an answer before the reproduction claim is final.

- 🔴 **CUPMem session mode: default** `relevant_only` **ingests only gold sessions. Which mode gave 68%?**
  `relevant_only` skips the 48 distractors, which makes the task far easier. The reported **$0.37/instance** at $0.15/M input implies 2M+ tokens, too many for 2 sessions, so it points to `full` . Plan: run both, report `relevant_only` as an oracle ablation, ask the authors. Possibly the headline finding.
- 🔴 **Target-model temperature not stated in the paper**
  The official code sends none, so OpenAI's default of 1.0 applies. We use 0 (deviation D3).
- 🔴 **Judge errors silently scored as FAIL in official code**
  A failed judge call lowers the score and leaves no trace. We record them instead (deviation D2).
- 🔴 **KEEP/STALE/REPLACE/UNKNOWN (§5) vs ADD/REFINE/REPLACE/NO_OP (App. F.2)**
  The paper describes CUPMem's decision labels two different ways. Settle it against `cup_mem/write/` before building M.
- 🔴 **Embedding model** `all-MiniLM-L6-v2` **only in README**
  The embedder shapes what retrieval finds. M has to use the same one, and the paper does not say so.
- 🟠 **Find and read *The Memory Trust Gap***
  Cited in the blueprint but not yet located. Also check papers citing STALE.

10 Deviations logged

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Every place we differ from the official code, with the reason. This list becomes the paper's reproducibility statement.

- 🟢 **D1 truncation RNG seeded by (seed, uid, dim)**
  The official code uses unseeded `random.choice` , so two runs can drop different sessions. Seeding makes reruns exact.
- 🟢 **D2 judge errors recorded and re-judged on resume**
  Stored in `Verdict.error` , retried on resume, and counted in the run summary instead of becoming silent FAILs.
- 🔴 **D3 target temperature 0 (official: API default)**
  Needed for determinism and caching. Still to decide: also run at the API default for exact E1 fidelity?
- 🟢 **D4 tiktoken + o200k fallback, same as official**
  Listed for completeness: no real deviation. Token counts match the API within 2 tokens.

11 Timeline

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A 13-week plan tied to UTokyo PhD applications.

- **13-week blueprint, started 28 Sep 2026**
  Runs to the end of December, in line with the arXiv target.
- 🟢 **Week 1: artifacts found, data verified, harness built**
  Finding the public code and data turned RQ1 into a three-way comparison.
- 🟢 **3 Oct: first API runs, B0 smoke test**
  Confirmed truncation, rate limits and cost on 4 instances before scaling to the 50-instance dev run.
- 🟠 **Nov 2026: email UTokyo professors**
  Aim to have RQ1 numbers to show by then.
- 🟠 **Late Dec 2026: post to arXiv**
  Then submit to MLRC / TMLR or an ACL/EMNLP workshop.

`Sources: notes/reproducibility_log.md · README.md · configs/`

`Built from notes/mindmap_panels.html`